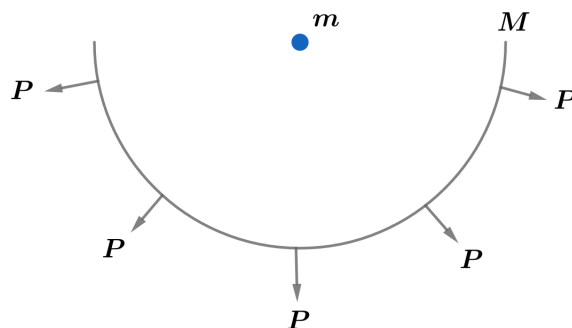


2018B F=ma Exam: Problem 24

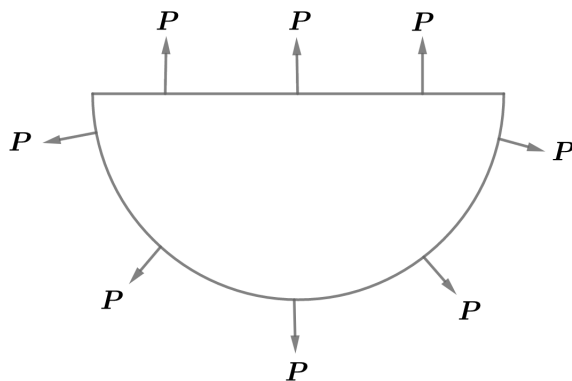
Kevin S. Huang



The gravitational force between the particle and each part of the shell is the same so the shell has the same pressure over its surface. Flipping the direction of gravity leaves the same magnitude but now the shell effectively contains a fluid:

$$P = \frac{F_{tot}}{A_{tot}} = \frac{GMm}{R^2(4\pi R^2)} = \frac{G\sigma m}{R^2}$$

By Pascal's principle, if the weight of the fluid is negligible, then the pressure throughout an enclosed volume is the same. For convenience, consider adding a lid to the shell to enclose the fluid:



Since the isolated system stays at rest, the downward force on the bottom is equal to the upward force on the top:

$$F_{bot} = F_{top} = PA = \frac{G\sigma m}{R^2} \pi R^2 = \pi G\sigma m$$

Hence, in the original scenario, the magnitude of the force is $F = \pi\sigma m$ and since gravity is attractive, it is downward on the particle. Thus, the answer is E.